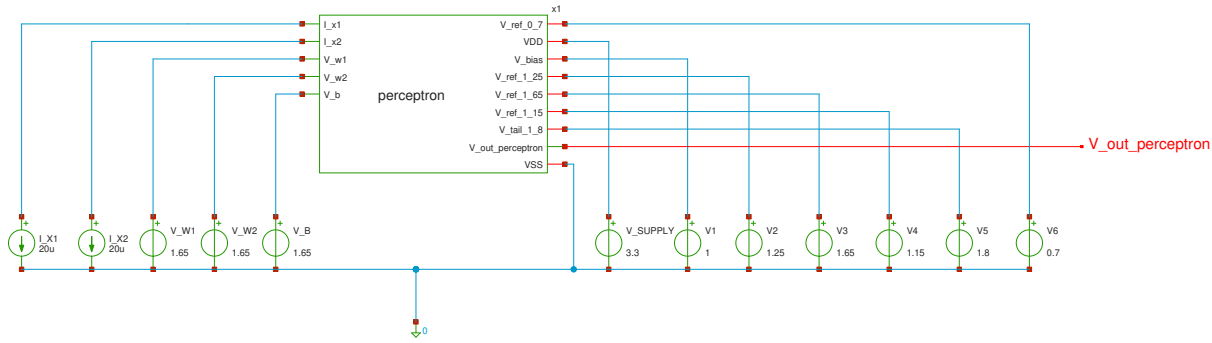


analog-perceptron-gf180mcu/schematics/tb_perceptron.sch

MODELS

```
.include $::180MCU_MODELS/design.ngspice
.lib $::180MCU_MODELS/sml41064.ngspice typical
.lib $::180MCU_MODELS/sml41064.ngspice res_typical
* .lib $::180MCU_MODELS/sml41064.ngspice res_statistical
```



COMMANDS

```
.control
* --- start from neutral ---
alter V_W1 = 1.65
alter V_W2 = 1.65
alter V_B = 1.65
alter I_X1 = 0
alter I_X2 = 0

* === Graph 1: Weight sweep (inhibitory / neutral / excitatory) ===
alter I_X2 = 0
alter V_W2 = 1.65
alter V_B = 1.65
let n = 0
dowhile n <= 2
  let vw1 = 1.45 + n * 0.2
  alter V_W1 = vw1
  dc I_X1 0 50u 1u
  wrdata /foss/designs/analog-perceptron-gf180mcu/plots/perceptron_graph1_weights.txt v(V_out_perceptron)
  set appendwrite
  let n = n + 1
end
unset appendwrite
alter V_W1 = 1.65

* === Graph 2: Sensor 1 sweep, Sensor 2 interference ===
alter V_W1 = 1.85
alter V_W2 = 1.85
alter V_B = 1.65
let n = 0
dowhile n <= 2
  let ix2 = n * 25u
  alter I_X2 = ix2
  dc I_X1 0 50u 1u
  wrdata /foss/designs/analog-perceptron-gf180mcu/plots/perceptron_graph2_sensors.txt v(V_out_perceptron)
  set appendwrite
  let n = n + 1
end
unset appendwrite
alter V_W1 = 1.65
alter V_W2 = 1.65
alter I_X2 = 0

* === Graph 3: Bias shifting ===
alter I_X2 = 0
alter V_W1 = 1.85
let n = 0
dowhile n <= 2
  let vb = 1.45 + n * 0.2
  alter V_B = vb
  dc I_X1 0 50u 1u
  wrdata /foss/designs/analog-perceptron-gf180mcu/plots/perceptron_graph3_bias.txt v(V_out_perceptron)
  set appendwrite
  let n = n + 1
end
unset appendwrite
alter V_W1 = 1.65
alter V_B = 1.65

* === 3D Dataset A: I_X1 x I_X2 -> V_out ===
alter V_W1 = 1.85
alter V_W2 = 1.85
alter V_B = 1.65
let n = 0
dowhile n <= 25
  let ix2b = n * 2u
  alter I_X2 = ix2b
  dc I_X1 0 50u 1u
  wrdata /foss/designs/analog-perceptron-gf180mcu/plots/3d_sensors.txt v(V_out_perceptron)
  set appendwrite
  let n = n + 1
end
unset appendwrite
alter V_W1 = 1.65
alter V_W2 = 1.65
alter I_X2 = 0

* === 3D Dataset B: V_W1 x V_W2 -> V_out ===
alter I_X1 = 50u
alter I_X2 = 50u
alter V_B = 1.65
let n = 0
dowhile n <= 20
  let vw2 = 1.45 + n * 0.02
  alter V_W2 = vw2
  dc V_W1 1.45 1.85 0.01
  wrdata /foss/designs/analog-perceptron-gf180mcu/plots/3d_weights.txt v(V_out_perceptron)
  set appendwrite
  let n = n + 1
end
unset appendwrite

* --- final state: neutral, zero input ---
alter V_W1 = 1.65
alter V_W2 = 1.65
alter V_B = 1.65
alter I_X1 = 0
alter I_X2 = 0
```